

SERENA COFANO

Software Engineer & Security Engineer · PhD in Cybersecurity

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SUMMARY

Software and security engineer with a PhD in Cybersecurity and 4+ years of combined industry and research experience. Specialized in software supply chain security, SBOM generation, and vulnerability assessment in Python ecosystems. Delivered code in Java, Python, C#, and Angular across enterprise and academic settings. Knowledge of CRA, NIS2, GDPR, and OWASP Top 10.

EXPERIENCE

Visiting PhD Researcher 10/2024 - 06/2025

KTH Royal Institute of Technology - Stockholm, Sweden

- Built a Java-based runtime dependency tracker enabling runtime transparency.
- Co-organized the 4th KTH Workshop on the Software Supply Chain.

PhD Researcher - Software Supply Chain Security 12/2022 - 11/2025

IMT School for Advanced Studies & University of Genoa - Italy

- Researched SBOM generation and vulnerability assessment in Python ecosystems; published 3 peer-reviewed papers.
- Built static and dynamic analysis tools in Java using code instrumentation and Java Agent techniques.

Cybersecurity Consultant 09/2023 - 11/2023

CINI (National Interuniversity Consortium for Informatics) - Genoa, Italy

- Performed security requirements analysis and risk assessment for IT-Alert, Italy's national emergency alert system.
- Designed a Security-by-Design pipeline and delivered developer training.

Junior Software Developer 05/2021 - 05/2022

Aizoon Consulting - Genoa, Italy

- Developed full-stack enterprise features in C#, JavaScript, AngularJS, and Angular 2+ in an agile client-facing team.
- Owned bug fixing, feature delivery, and ongoing software maintenance across multiple projects.

PUBLICATIONS

ArXiv 2026 - "Classport: Designing Runtime Dependency Introspection for Java."
Cofano, Williams, Sharma, Monperrus.

ACNS 2025 - "The Impact of SBOM Generators on Vulnerability Assessment in Python:
A Comparison and a Novel Approach." *Benedetti, Cofano, Brighente, Conti.*

IEEE TrustCom 2024 - "SBOM Generation Tools in the Python Ecosystem: an In-Detail
Analysis." *Cofano, Benedetti, Dell'Amico.*

SIGBOVIK 2025 - "UPPERCASE IS ALL YOU NEED."
Andersson, Baudry, Bobadilla et al. (incl. Cofano).

EDUCATION

PhD in Cybersecurity 2022 - 2025
IMT School for Advanced Studies & University of Genoa - Italy

Summer School - AI and Cybersecurity 09/2025
TU Wien - Vienna, Austria

MSc in Computer Engineering 2020 - 2022
University of Genoa - Italy

BSc in Biomedical Engineering 2016 - 2019
University of Genoa - Italy

SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, SQL, Bash
Frameworks: Angular, AngularJS, Maven, MongoDB, Git, Docker, Linux, Android
Security: SBOM, supply chain security, vulnerability assessment, OWASP Top 10,
ISO 27001, GDPR, NIS2, CRA, Security-by-Design
Techniques: Java Agents, code instrumentation, static & dynamic analysis

LANGUAGES

Italian (native) · English (C1) · French (B1) · German (A2) · Spanish (A2)